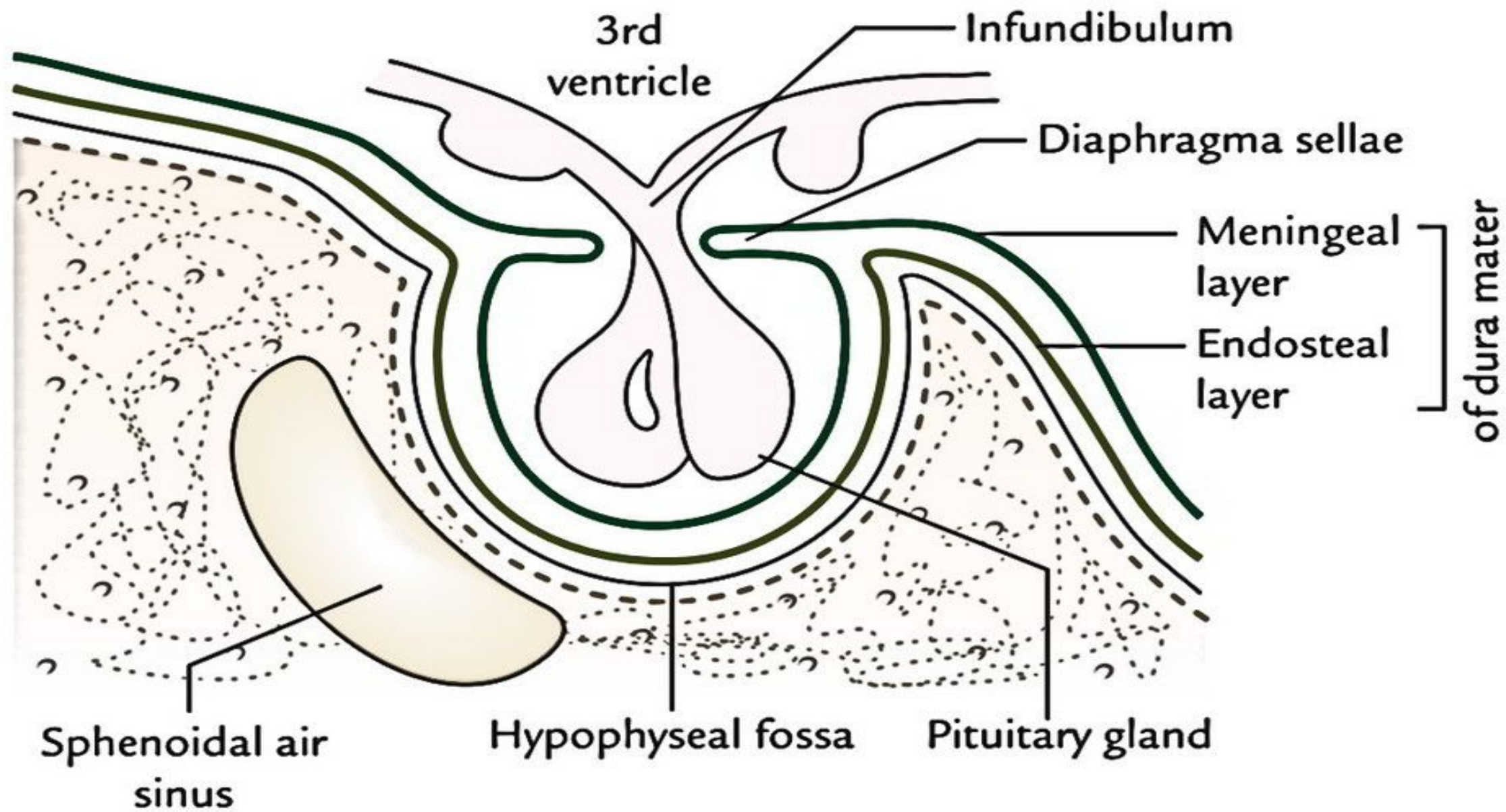


Pituitary Gland

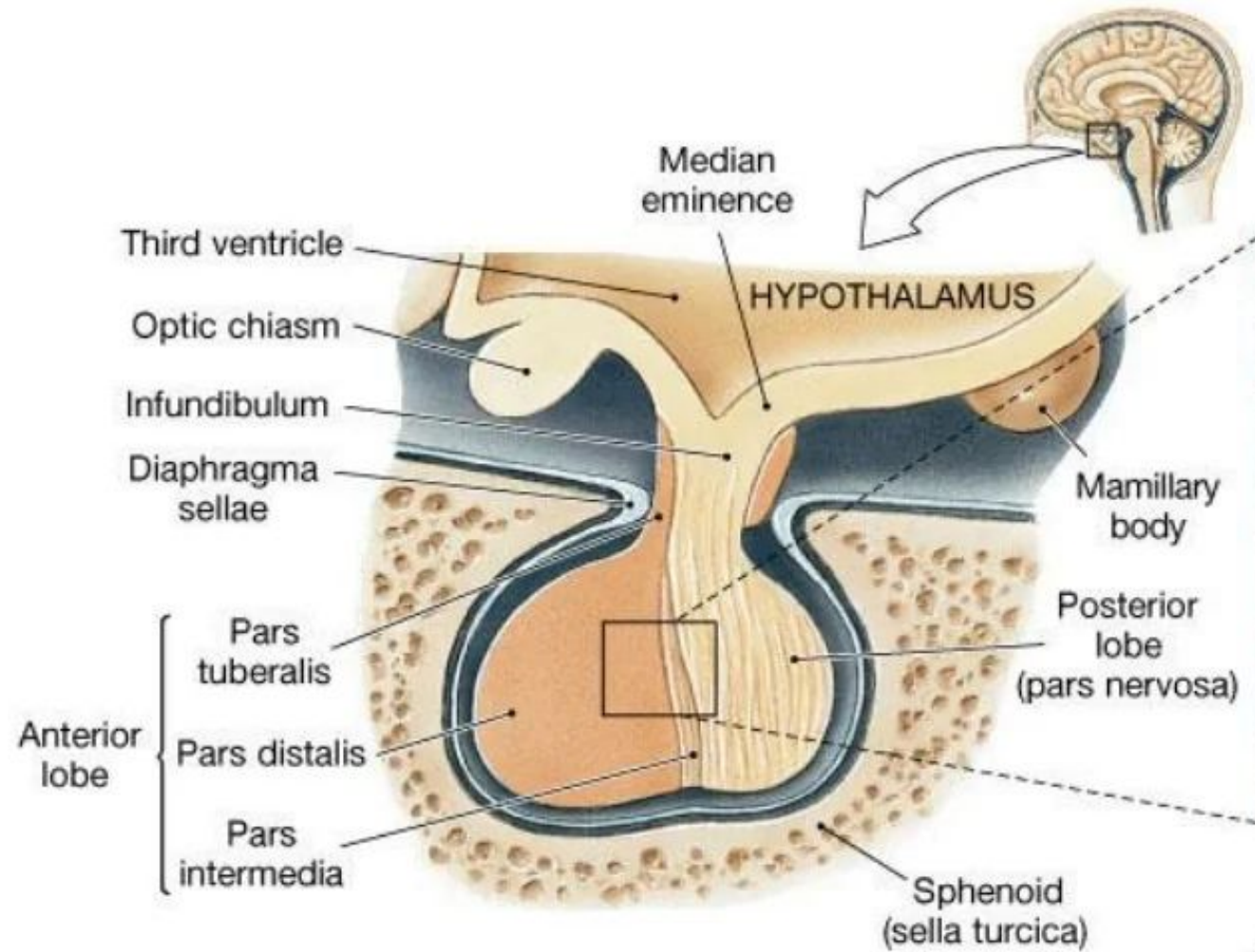
- The hypophysis cerebri is a small endocrine gland situated in the base of the brain.
- Master of gland because it produce a number of hormones which control the secretion of many endocrine glands of the body.
- Hypophyseal fossa or Sella turcica or Pituitary fossa.
- Fossa is roofed  diaphragma Sella
- The stalk of hypophysis cerebri pierces the diaphragma sellae and attach above to the floor of third ventricle.



Relation:

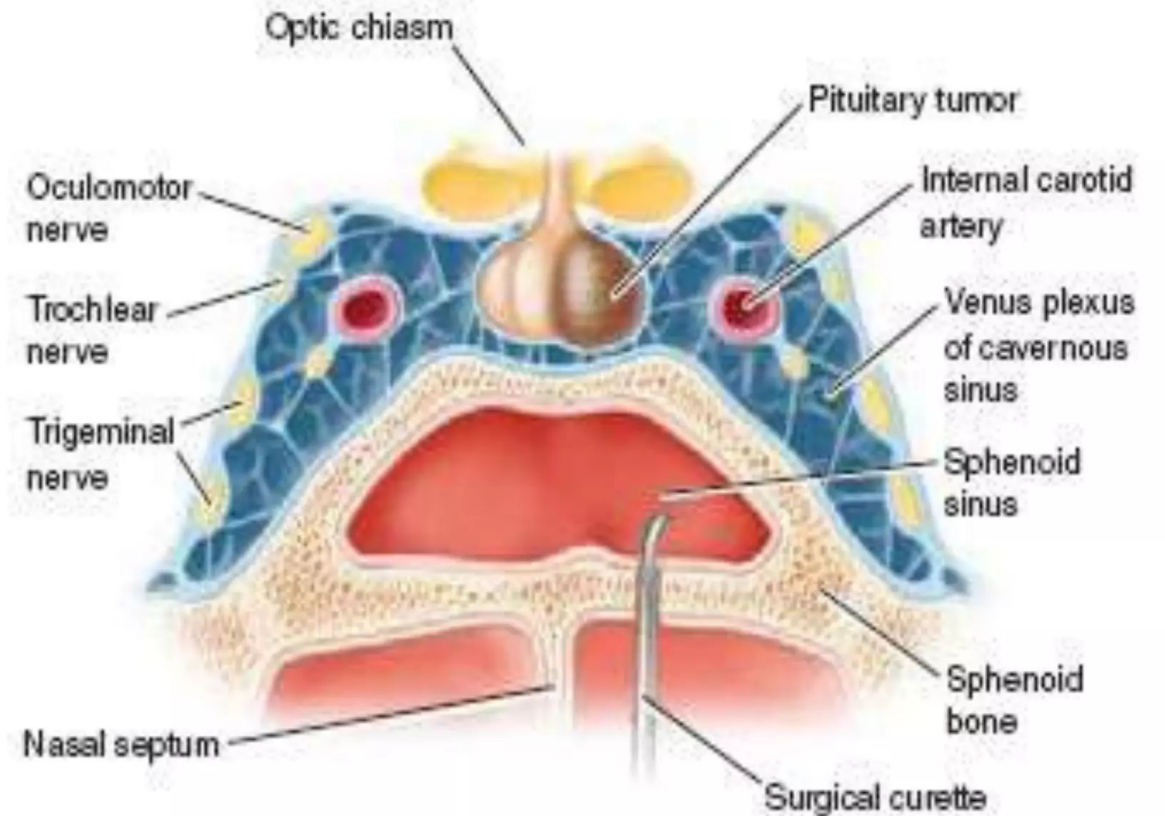
Superiorly:

1. Diaphragma sellae
2. Optic chiasma
3. Tuber cinerium
4. Infundibular recess of the third ventricle



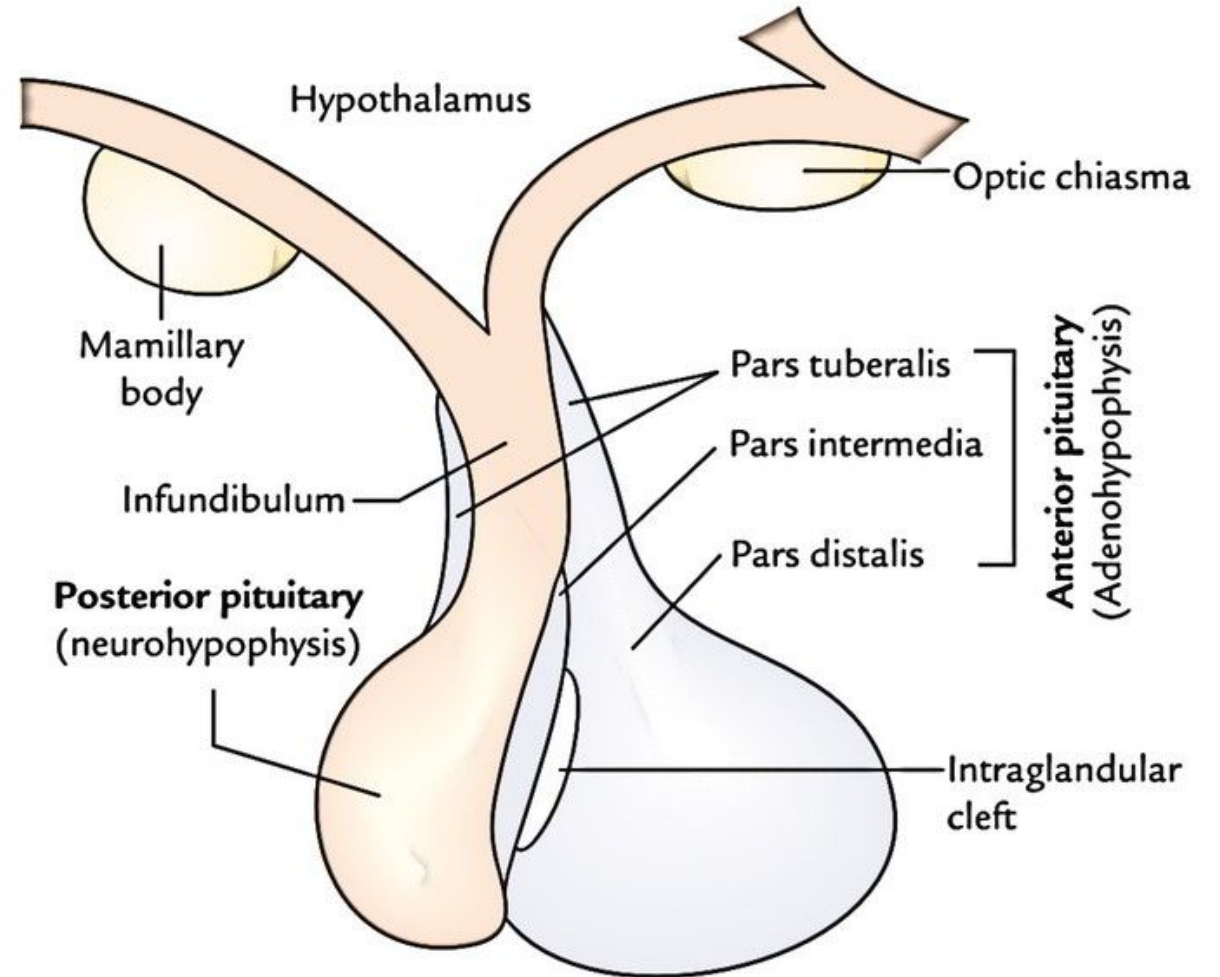
Inferiorly:

1. Irregular venous channels between the two layers of dura mater lining the floor of the hypophyseal fossa.
2. Hypophyseal fossa.
3. Sphenoidal air sinuses.

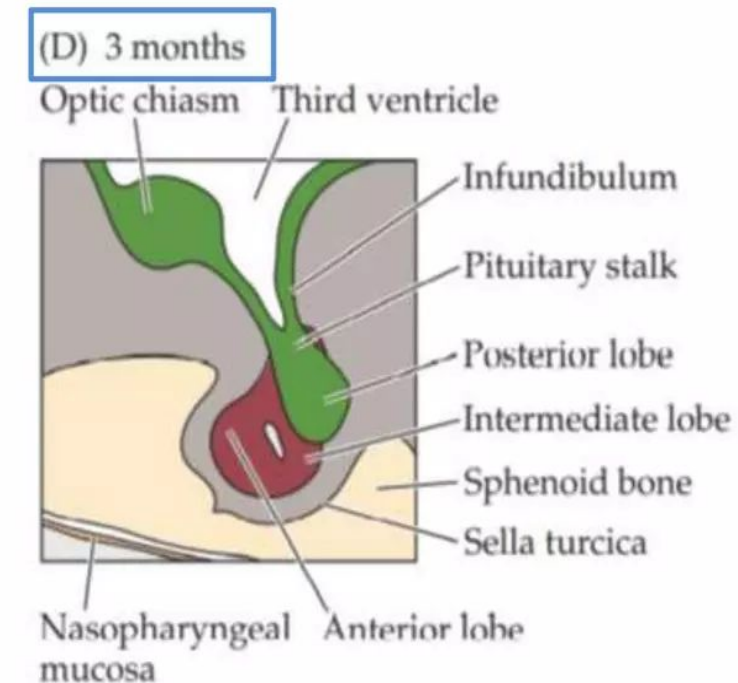
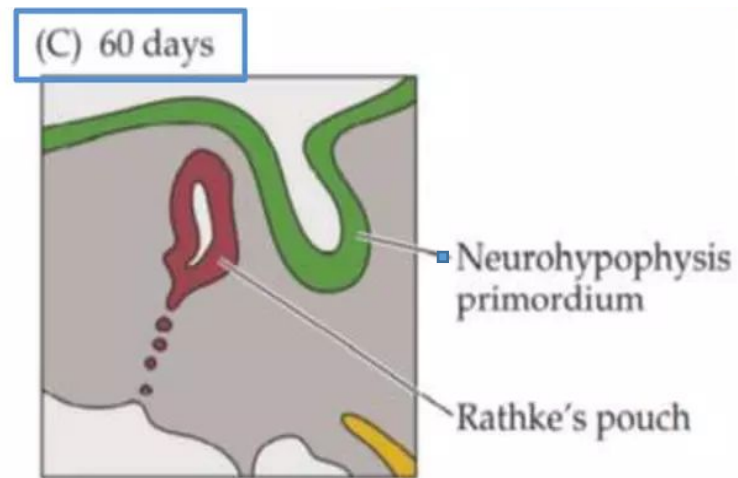
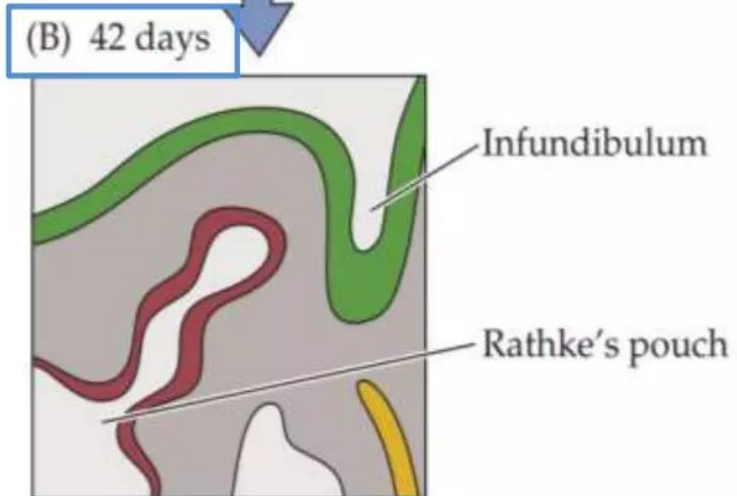
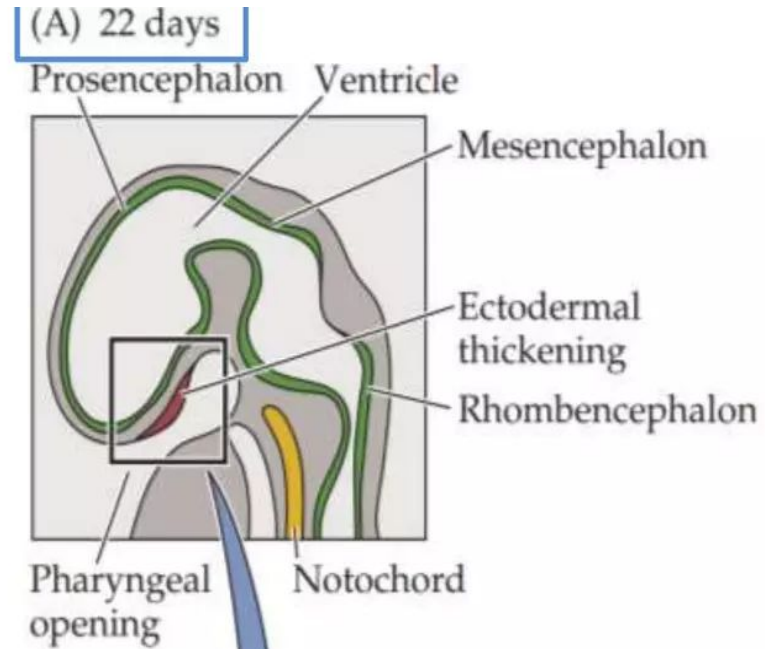


Parts/ Subdivision:

- Adenohypophysis
- Neurohypophysis



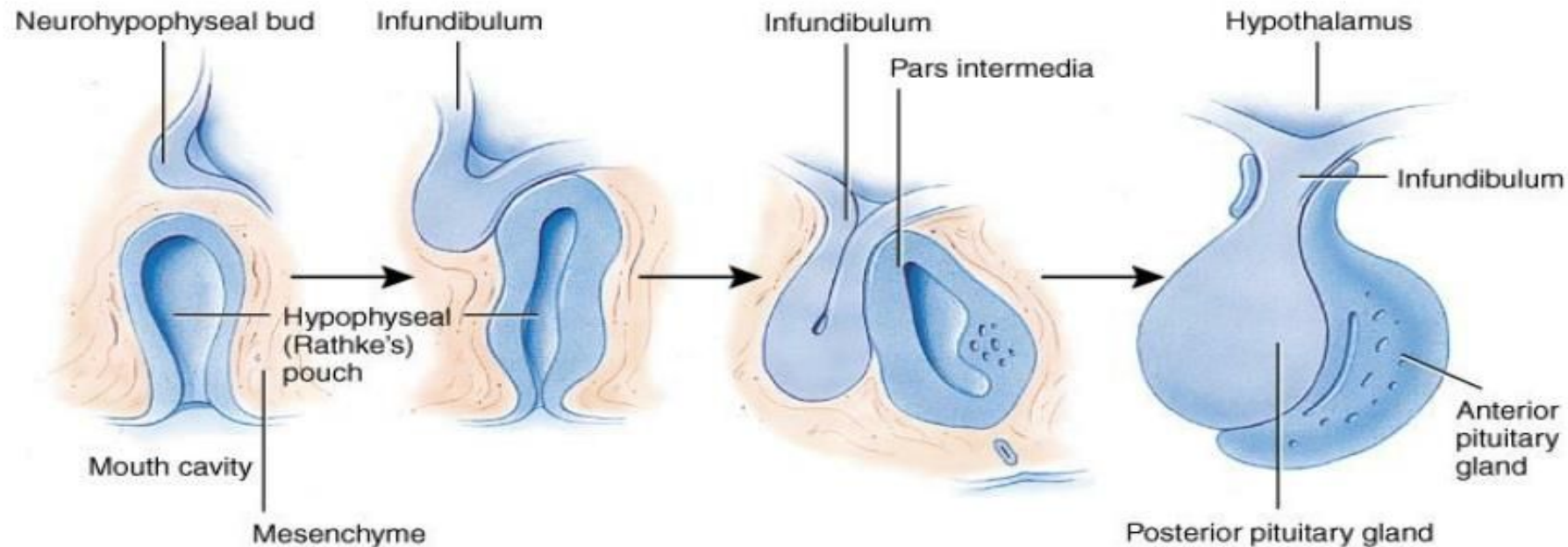
Development:



Adenohypophysis

Neurohypophysis

Development of Pituitary Gland



- Events occurring between 5 and 16 weeks of age

Arterial Supply:

- **Superior hypophyseal artery**
- **Inferior hypophyseal artery**

HISTOLOGY OF THE PITUITARY GLAND (HYPOPHYSIS)

Comprehensive Architectural & Cellular Organization SmartArt Diagram

PITUITARY GLAND (HYPOPHYSIS)

Master Endocrine Gland • Dual Embryological Origin

ANTERIOR LOBE (Adenohypophysis)

Epithelial Origin • Derived from Oral Ectoderm (Rathke's Pouch)

Pars Distalis

75% of gland mass. Cords of parenchymal cells & fenestrated sinusoids.

Pars Tuberalis

Vascularized collar around infundibular stalk. Highly vascular.

Pars Intermedia

Thin zone with basophilic cords & colloid-filled cysts (Rathke's remnants).

Pars Distalis Cell Classification (Staining Affinity)

Chromophils (50%) — Granulated

Acidophils (40%)

- **Somatotropes (50%):** GH
- **Lactotropes (15-20%):** Prolactin

Basophils (10%)

- **Corticotropes (15-20%):** ACTH
- **Thyrotropes (5%):** TSH
- **Gonadotropes (10%):** FSH, LH

Chromophobes (50%)

Non-Granulated

- Pale, clear cytoplasm
- Depleted/resting cells or stem progenitor cells

POSTERIOR LOBE (Neurohypophysis)

Neural Origin • Ventral Evagination of Diencephalon

Infundibulum

Connecting stalk containing hypothalamic-hypophyseal tract.

Pars Nervosa

Neurosecretory organ storing ADH and Oxytocin.

Pars Nervosa Structural Components

1. Unmyelinated Axons:

Axons of neurosecretory neurons whose cell bodies reside in hypothalamic nuclei (supraoptic & paraventricular).

2. Herring Bodies:

Eosinophilic axonal swellings containing neurosecretory vesicles storing **Oxytocin** and **Vasopressin (ADH)** with neurophysins.

3. Pituicytes:

Specialized glial support cells (GFAP-positive) with branching processes surrounding axons.

4. Fenestrated Capillary Network:

Allows rapid entry of released neurohormones into systemic circulation.

PARS DISTALIS (ANTERIOR PITUITARY)

Histological Classification: Staining Properties vs. Functional Activity

PARS DISTALIS PARENCHYMA

Accounts for ~75% of the hypophysis • Arranged in cords & clusters supported by fenestrated capillaries

1. CLASSIFICATION BY STAINING PROPERTIES

A. Chromophils (50% of total cells) — Stain intensely due to secretory granules

Acidophils (~40%)

Eosinophilic

- Granules have affinity for **acidic dyes** (stain orange/red).
- Cytoplasm is packed with proteinaceous hormone granules.
- Includes **Somatotrophs** and **Lactotrophs**.

Basophils (~10%)

Basophilic / PAS+

- Granules have affinity for **basic dyes** (stain bluish).
- PAS-positive due to glycoprotein hormone content.
- Includes **Corticotrophs**, **Thyrotrophs**, **Gonadotrophs**.

B. Chromophobes (50% of total cells) — Non-staining

Chromophobes (~50%)

Clear / Pale

- Cytoplasm is devoid of secretory granules; stain poorly.
- Represent degranulated/resting chromophils or progenitor stem cells.

2. CLASSIFICATION BY FUNCTIONAL ACTIVITY

Categorized into **5 distinct functional cell types** based on immunocytochemistry and secreted hormones:

Cell Type	Pop. (%)	Stain Profile	Hormones & Function
Somatotrophs	~50%	Acidophilic	Growth Hormone (GH): Promotes body/tissue growth.
Lactotrophs	15–20%	Acidophilic	Prolactin: Promotes mammary gland growth & lactation.
Corticotrophs	15–20%	Basophilic	ACTH & β-lipotropin: Adrenal cortex stimulation & lipolysis.
Gonadotrophs	~10%	Basophilic	FSH & LH: Ovarian follicle growth, ovulation & spermatogenesis.
Thyrotrophs	~5%	~5% / Basophilic	TSH (Thyrotropin): Stimulates thyroid hormone release.

Summary Correlation:

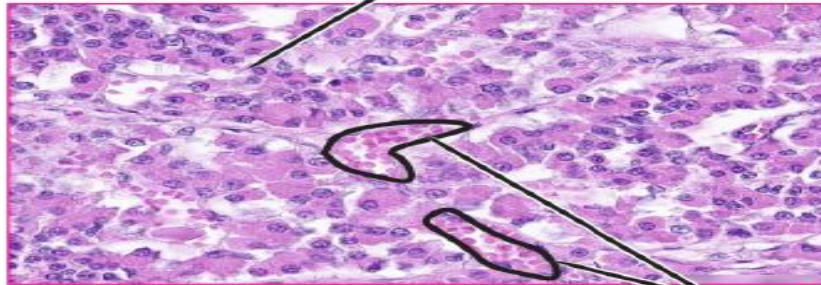
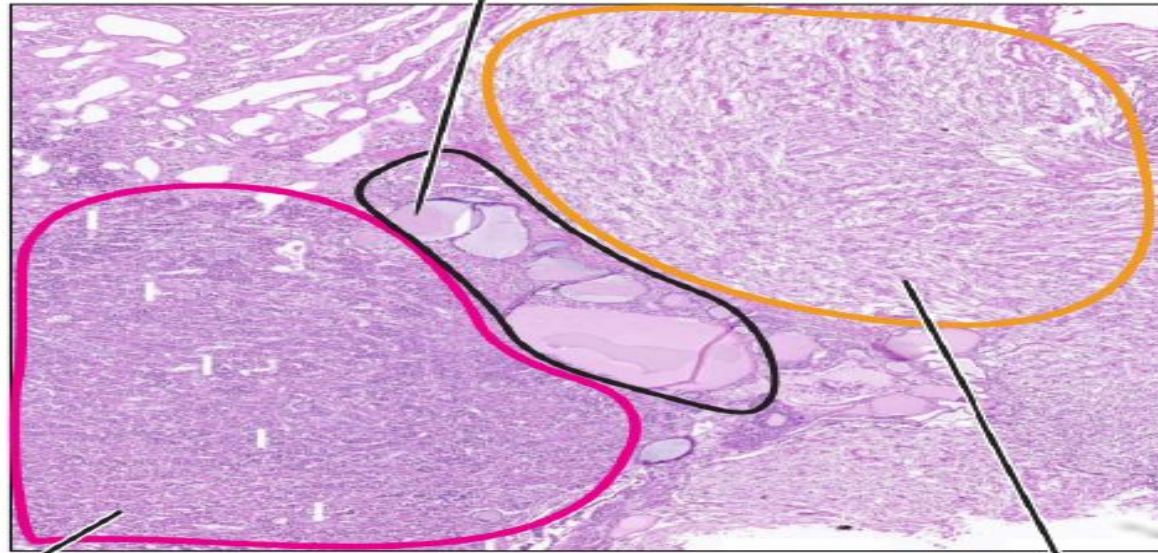
- **Acidophils (2 types):** Somatotrophs + Lactotrophs (~70% of chromophils)
- **Basophils (3 types):** Corticotrophs + Gonadotrophs + Thyrotrophs (~30% of chromophils)

Pituitary Gland

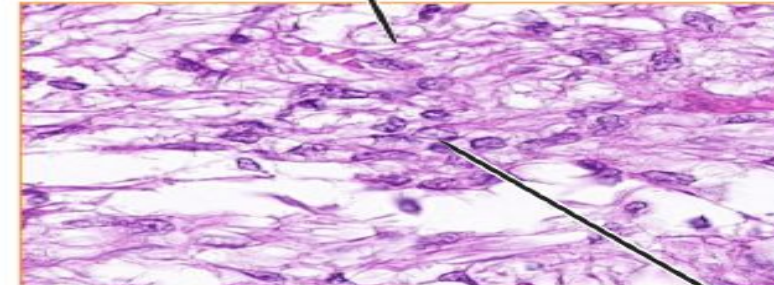
Anterior Pit.
(adenohypophysis)

**pars
intermedia**
(w/ colloid)

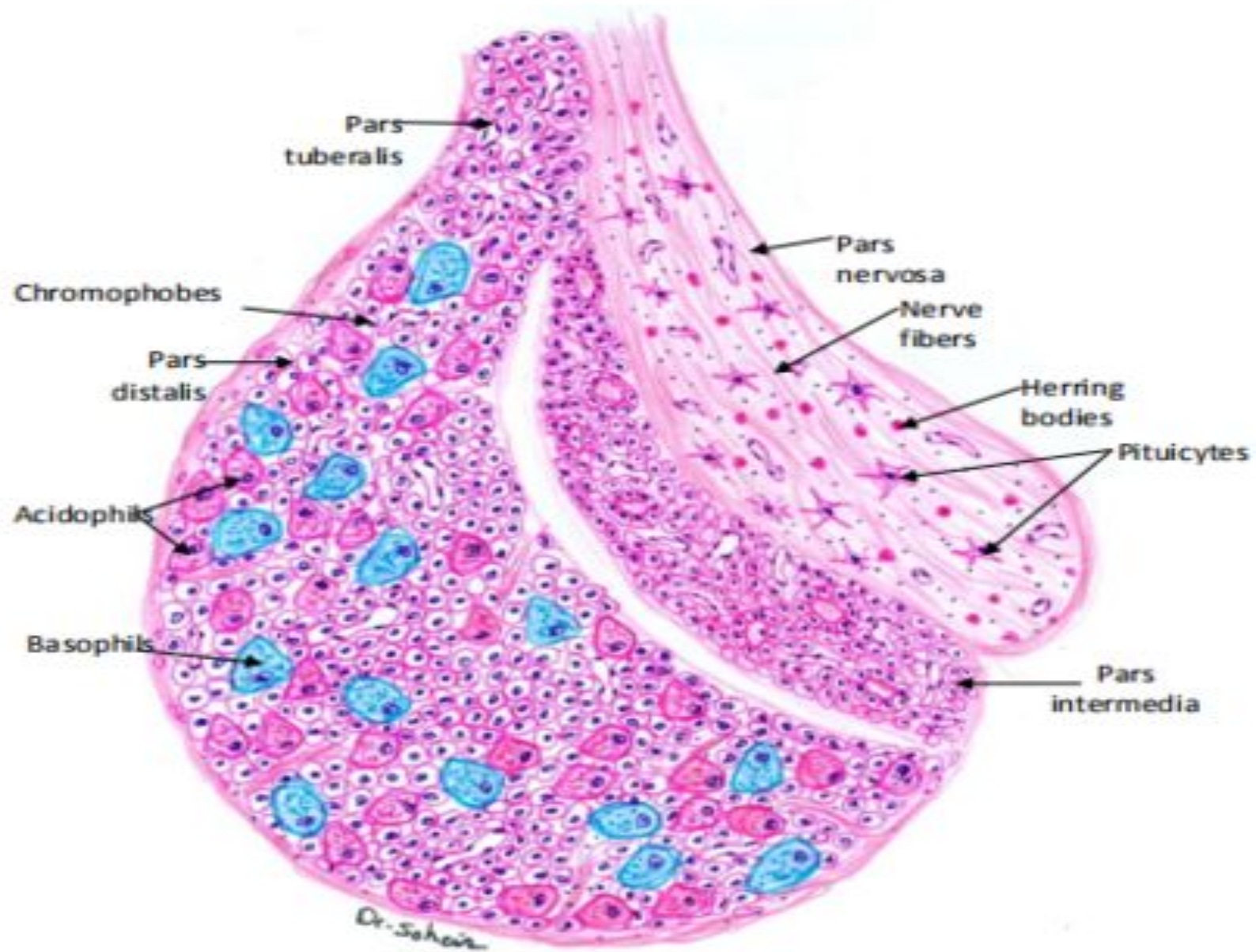
Posterior Pit.
(neurohypophysis)



Acidophilic & basophilic
hormone-secreting cells Sinusoids



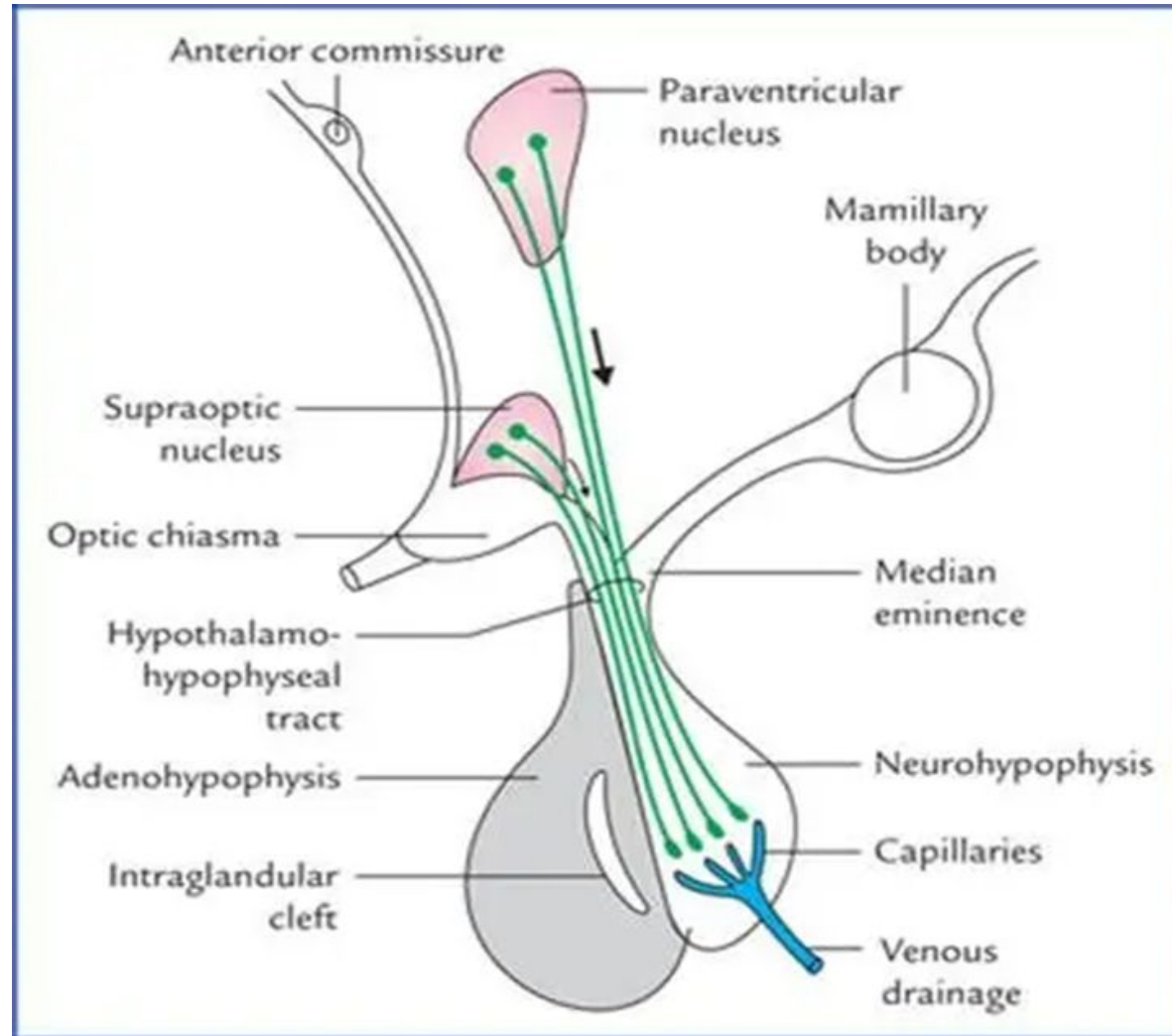
Unmyelinated
axons Nuclei of
pituicytes



Somatropes	Oval cells containing a centrally located spherical nucleus.	In electron microscope the cytoplasm of these cells contain a large number of secretory granules that measure 300 to 400µm in diameter.	50%	Acidophilic	Somatotropes secretes growth hormone (somatotropin)
Lactotropes	Polygonal cells with oval nuclei	600nm	15 to 20%	Acidophilic	Lactotropes secrete prolactin.
Corticotropes	Polygonal cells each having round nucleus located in a eccentric position	250-400nm	15 to 20 %	basophilic	Secretes ACTH & β lipotropin

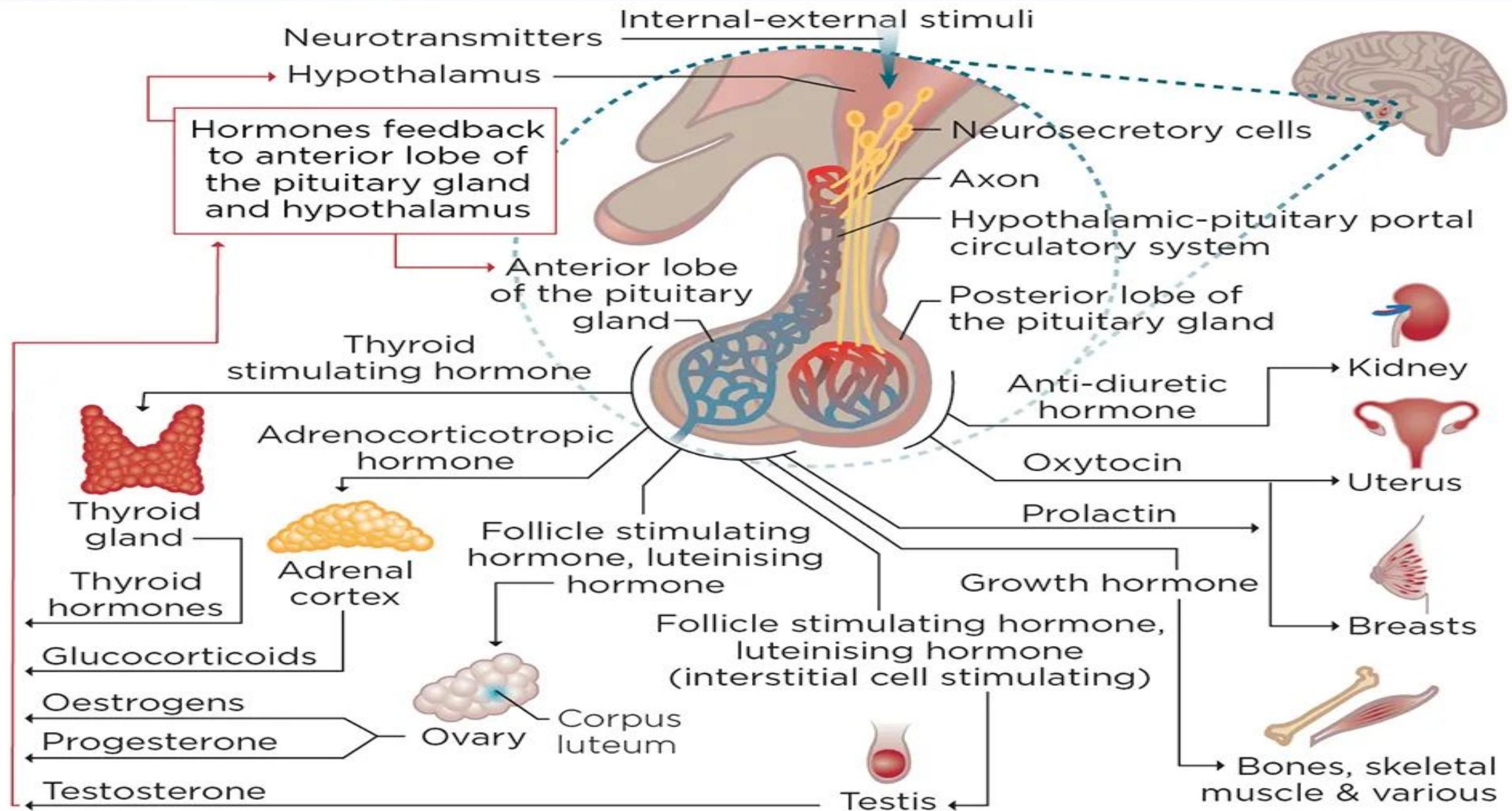
Gonadotropes	Oval cells with round nuclei	200- 400nm	10%	Basophilic	Secretes FSH & LH
Thyrotropes	Relatively large, oval cells with round , eccentrically placed nuclei.	150nm	5%	Basophilic	Secretes TSH (thyrotrophs)

Hypothalamohypophyseal tract:



- The hormone vasopressin and oxytocin are synthesized in the nerve cell of the supraoptic and paraventricular nuclei of the hypothalamus.
- The hormones are passed along the axons together with carrier proteins called neurophysins and are released at axon terminals.
- Here the hormones are absorbed into the bloodstream in fenestrated capillaries of the posterior lobe of pituitary.

Fig 3. **Hormones of the posterior and anterior pituitary**

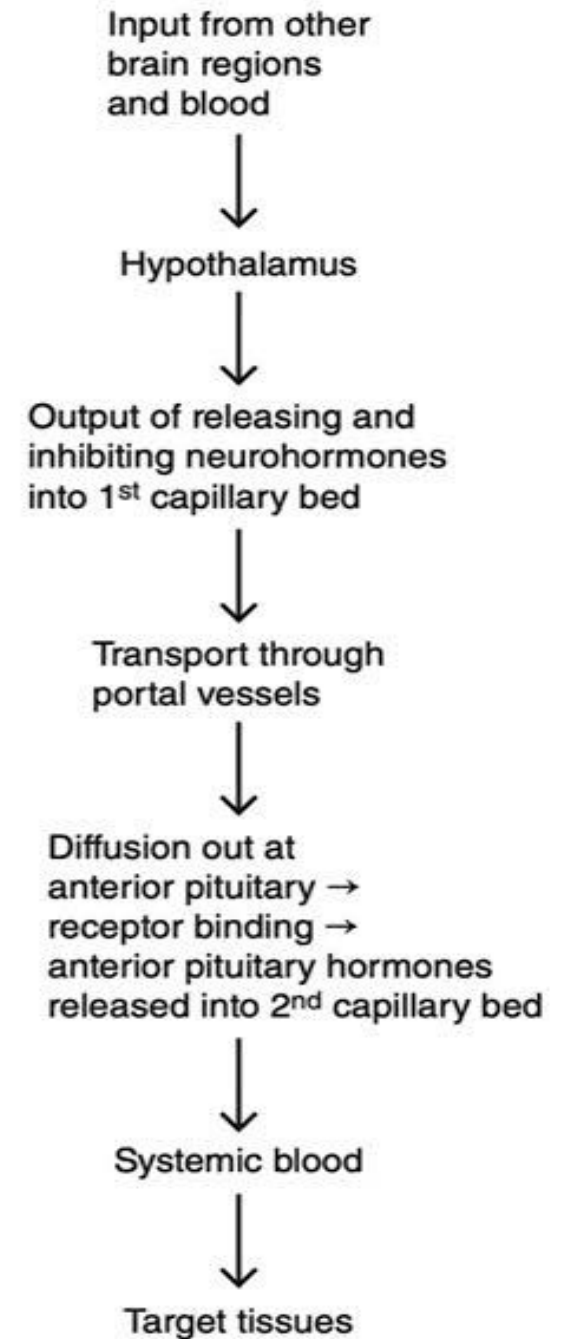
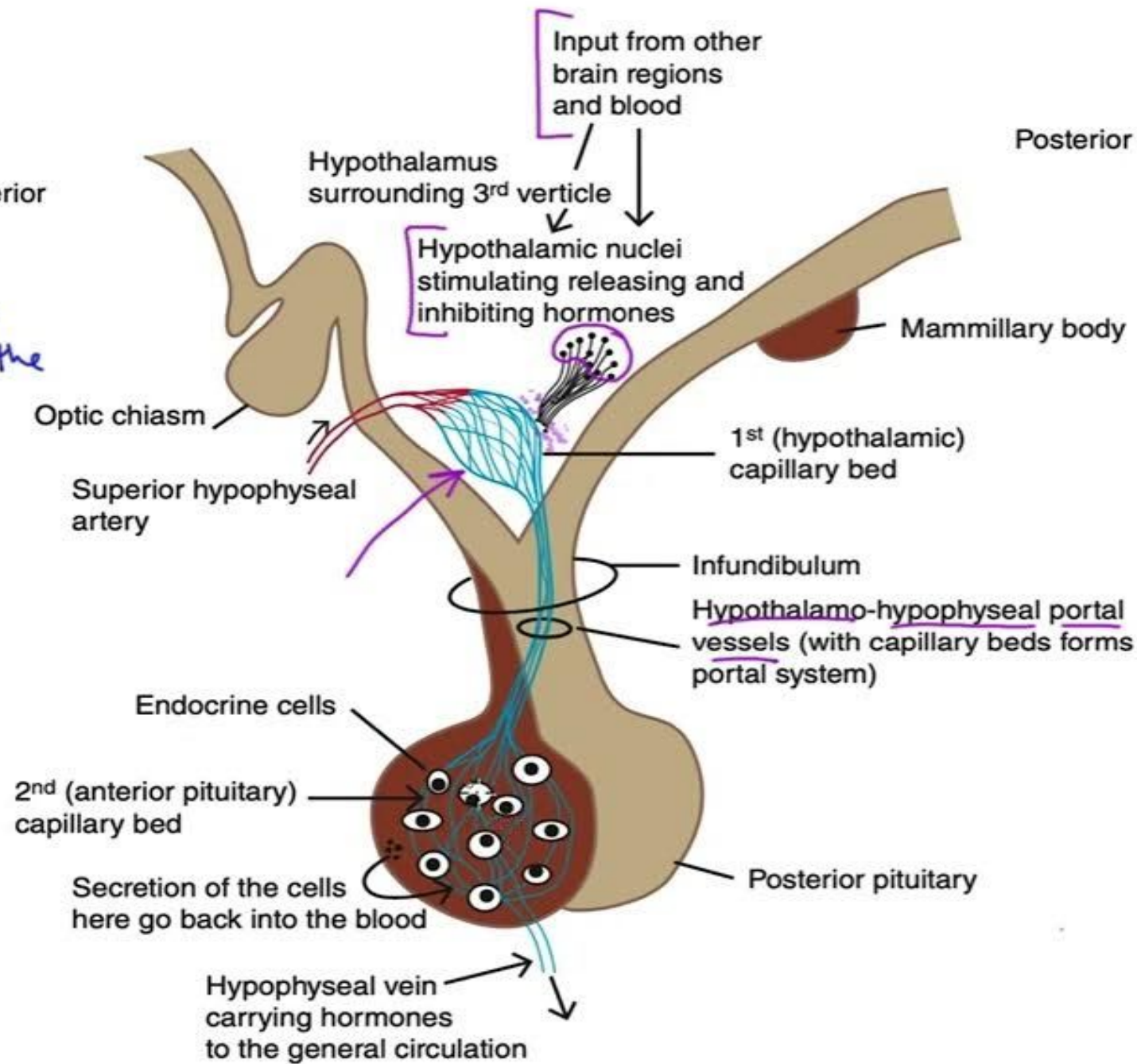


Hypophyseal portal system:

- Neurosecretory cell situated mainly in the medial zone of the hypothalamus are responsible for the production of the releasing hormone and release-inhibitory hormones.
- The hormones are packed into granules and transported along the axons of these cells into the medial eminence and infundibulum.
- Here, granules are released by exocytosis onto the fenestrated capillaries at the upper end of the hypophyseal portal system.

B

Hypothalamic control of the anterior pituitary



- The hypophyseal portal system is formed on each side from superior hypophyseal artery, which is the branch of internal carotid artery.
- The artery enters the medial eminence and divides into tufts of capillaries.
- These capillaries drain into long and short descending vessels that end in the anterior lobe of the hypophysis by dividing into vascular sinusoids that pass between the secretory cells of the anterior lobe.
- The portal system carries the releasing hormones and inhibiting hormones to the secretory cells of the anterior lobe of pituitary.

THANK YOU